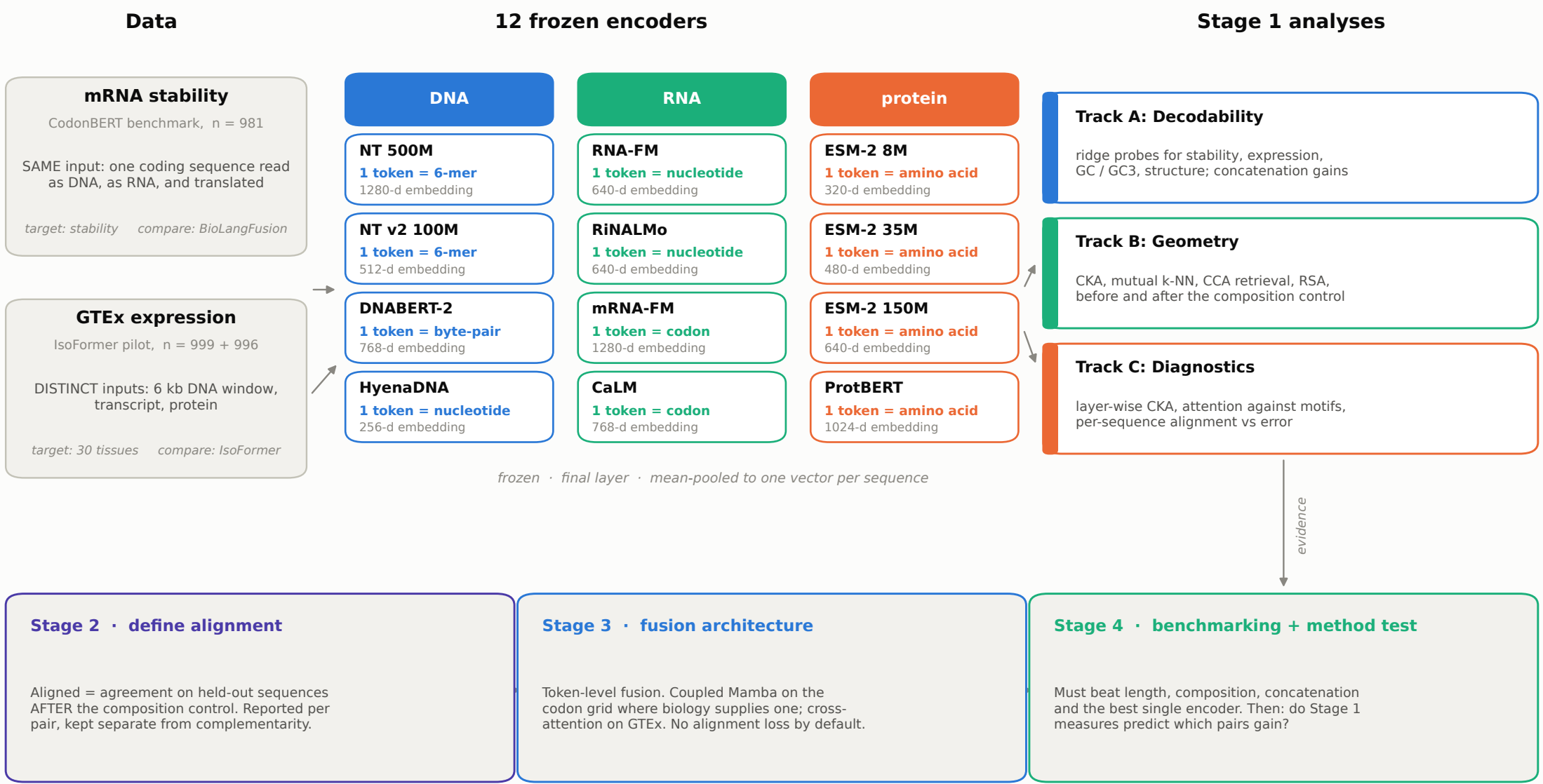
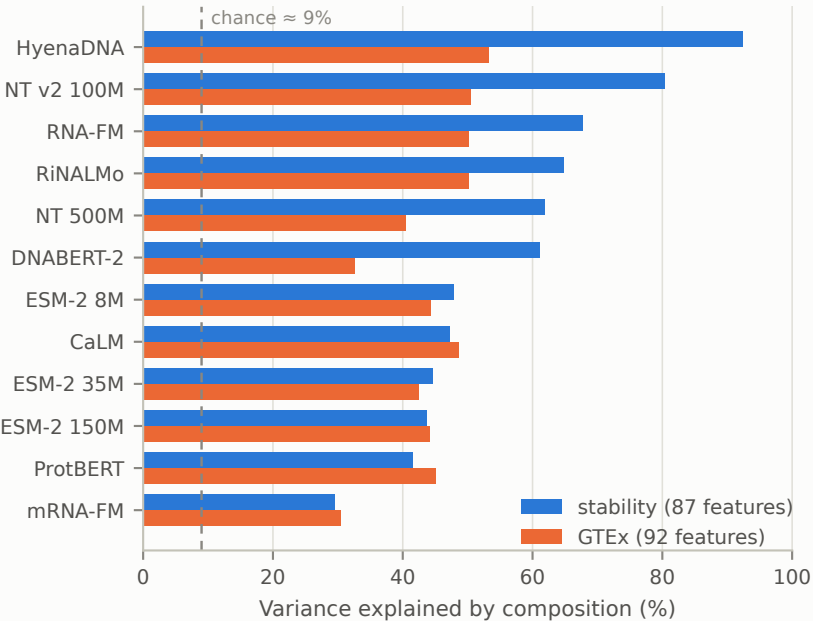


Choosing and combining biological foundation models: Stage 1 design

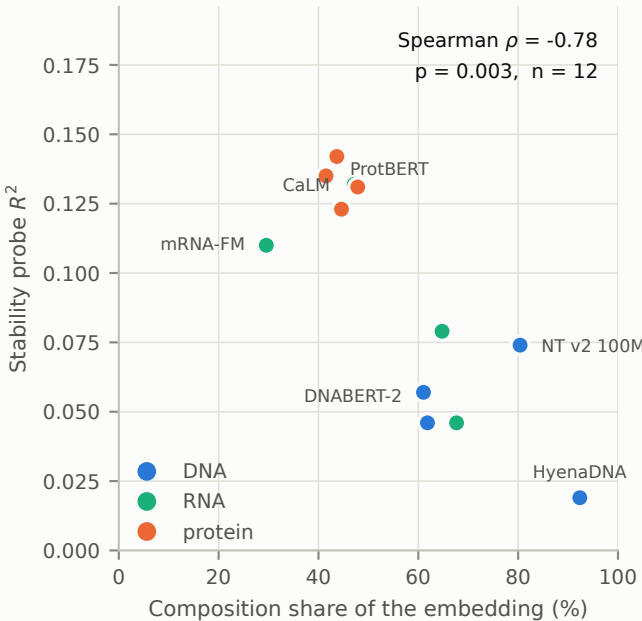


What sequence composition alone explains

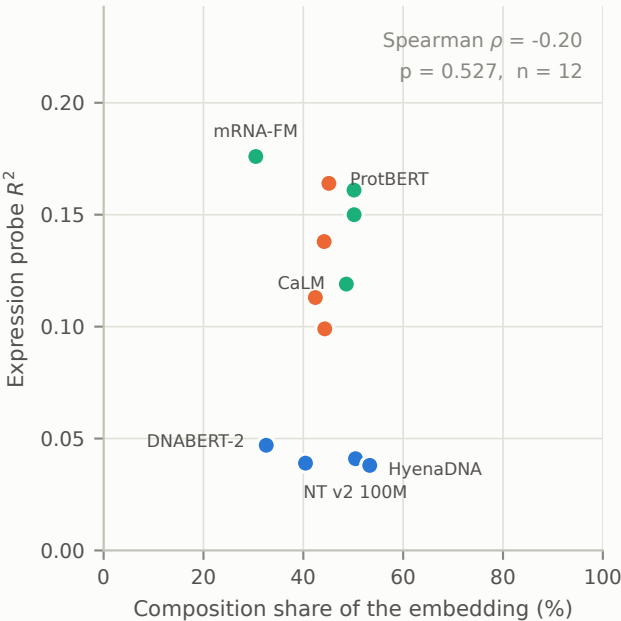
a How much of each embedding is composition



b Stability: less composition, better probe

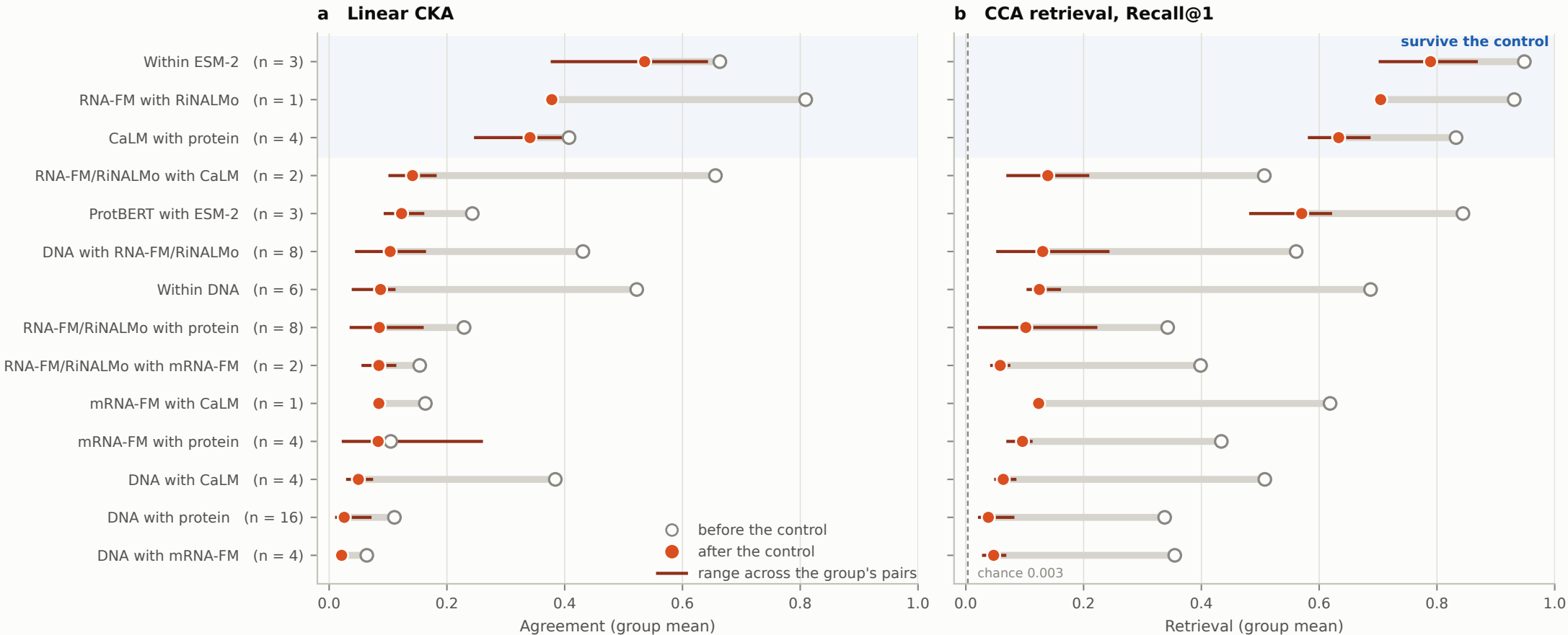


c GTEx: no such relation

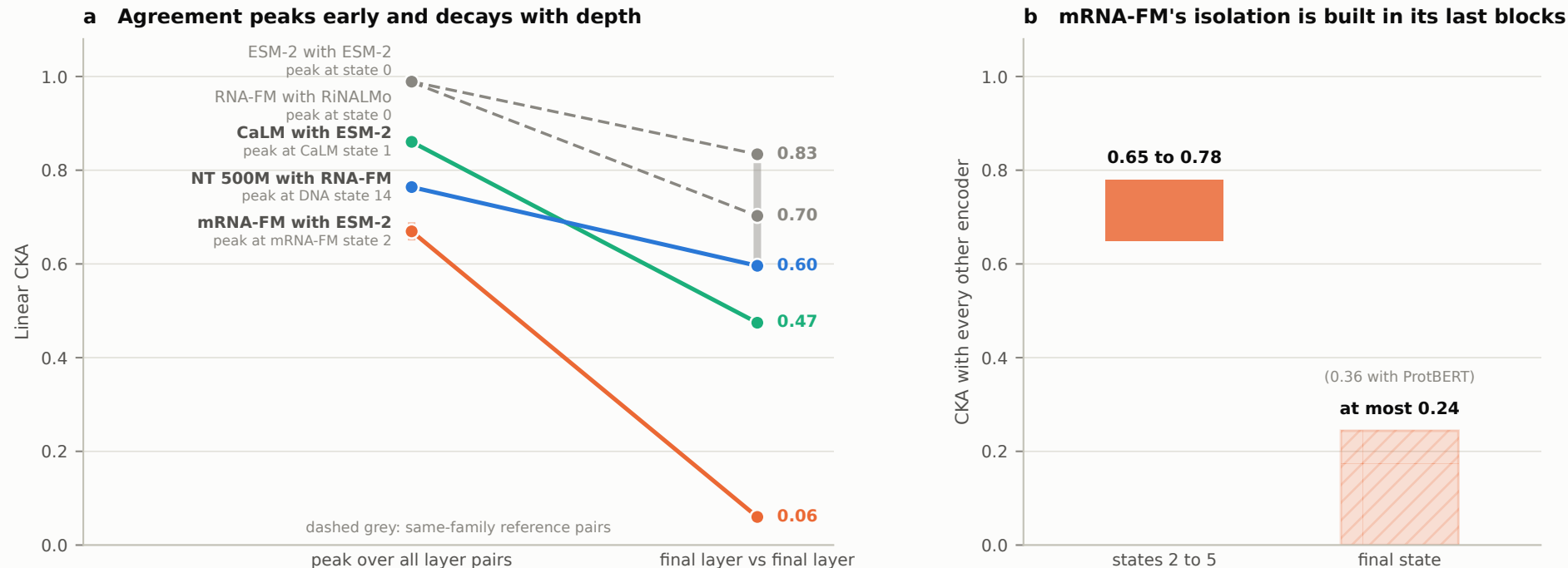


Composition = GC, GC3, log length, 64 codon and 20 amino-acid frequencies (87 on stability; 92 on GTEx, which adds the DNA window and the untranslated regions). Each panel uses its own dataset's features: plotting GTEx R^2 against the stability shares instead gives $\rho = -0.63$, $p = 0.028$, but compares two different feature sets. R^2 is the mean over ten fold assignments.

Most cross-encoder agreement is sequence composition

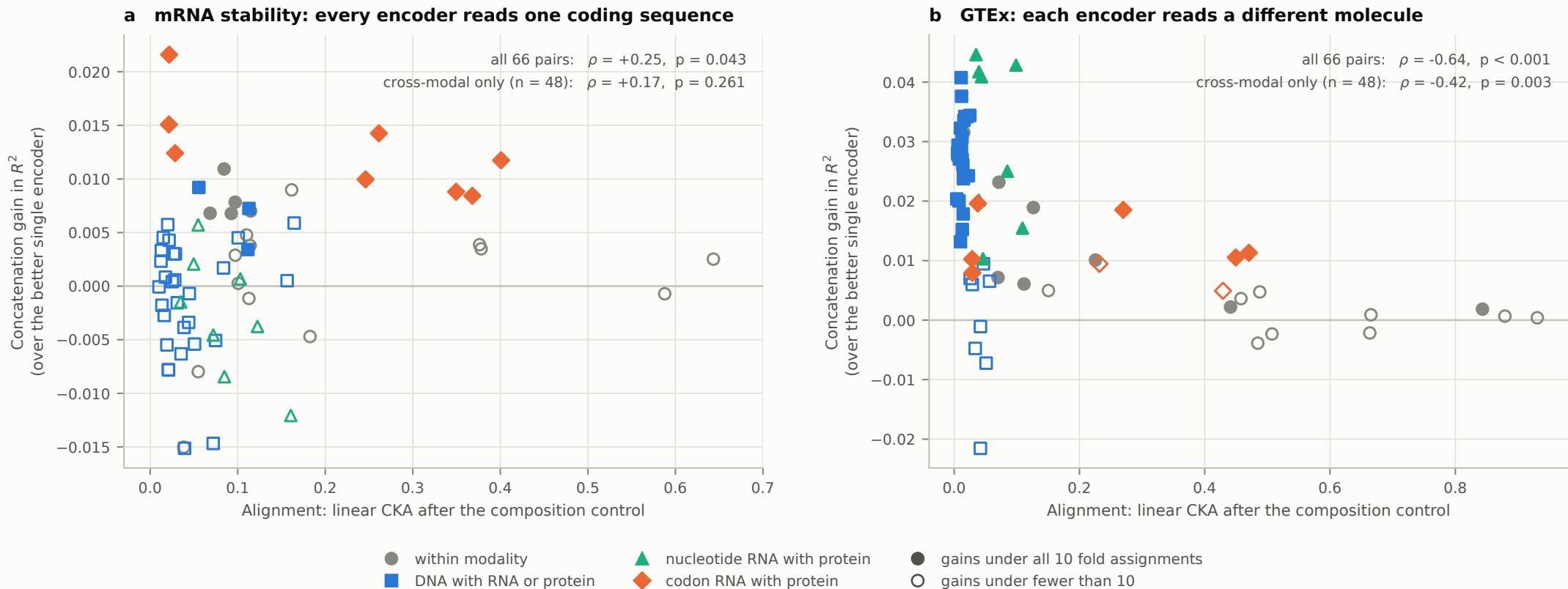


Where cross-encoder agreement lives inside the networks



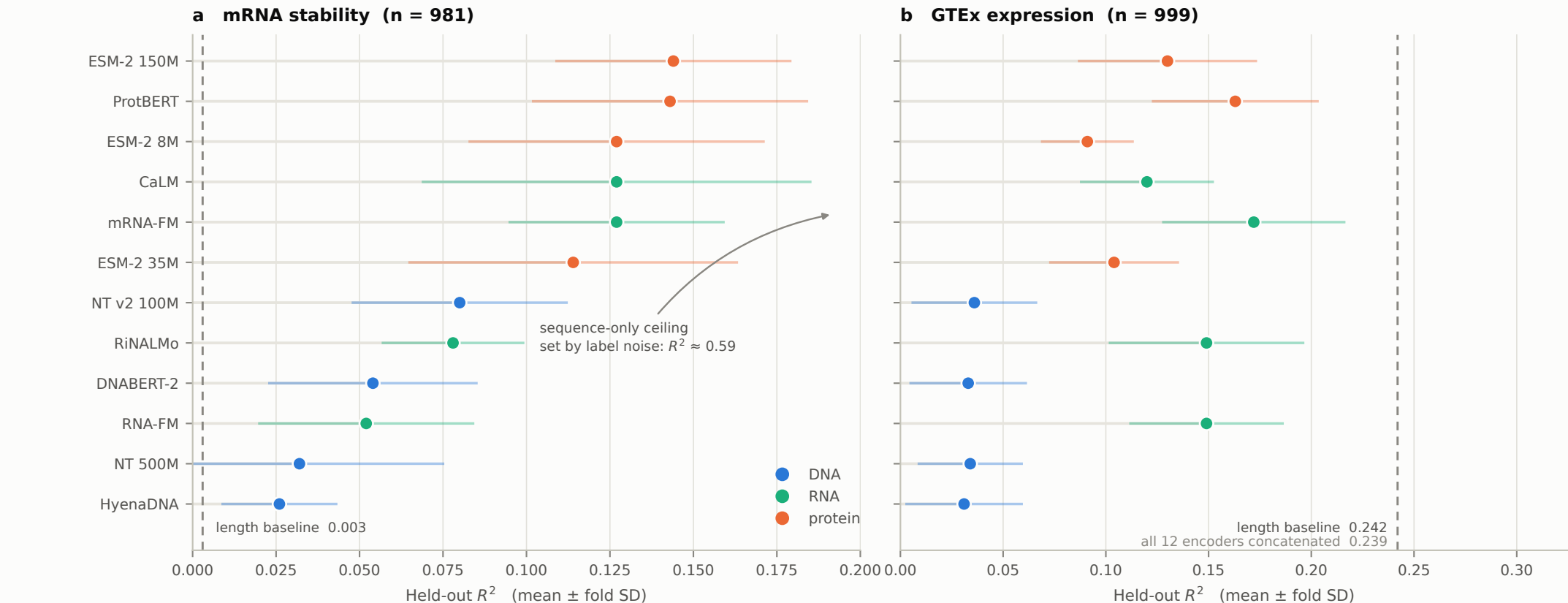
Layer-wise CKA on a 194-sequence subsample of the stability data, every hidden state mean-pooled. Linear CKA is inflated at small n: these final-layer values exceed the 981-sequence values behind Figure 3 for 63 of the 66 pairs, by up to 0.079, so read this figure within itself rather than against Figure 3. Shaded bars span the ESM-2 models where a row covers more than one pair.

Does alignment predict where combining encoders helps?



Gain is the mean over ten further fold assignments. On GTEx the pooled correlation is carried largely by the within-modality pairs ($\rho = -0.85$, $n = 18$), where two DNA encoders read the same window and are redundant by construction; the cross-modal half is the result that bears on the Stage 2 decision. No trend line is drawn, because Spearman is a rank statistic.

What a linear probe can read from each frozen encoder



Ridge probes on frozen mean-pooled embeddings; five folds that keep identical sequences (stability) or transcripts of one gene (GTEx) together. On stability, repeated sequences carry labels differing by a median SD of 0.49, which caps any sequence-only model near $R^2 = 0.59$, so 0.14 is about a quarter of what is reachable. Note the two panels use different x ranges.